

Modifier Shannon-Fischer Ratio

From Stochastic Chaos to Ordered Emission

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ABSTRACT

This paper introduces a novel computational framework for the real-time identification of emerging modal structures within high-entropy spectral environments. Traditional segmentation techniques often struggle to distinguish between stochastic noise patterns and the early stages of deterministic modal formation. We propose the Modified Shannon-Fisher Ratio (MSFR), an information-theoretic index designed to analyze Singularity Complexity. By evaluating the dynamic balance between global uncertainty (Shannon Entropy) and local structural sharpness (Fisher Information), the MSFR identifies the precise threshold where chaotic states transition into ordered emissions. This approach enables reconstructing deterministic modal patterns even when they are submerged within high-entropy backgrounds [1..8].

KEYWORDS: Information Theory, Spectral Waterfall Analysis, Real-time surrogate for computationally complexity metrics

MSFR SYNTHESIS

The synthesis of the Modified Shannon-Fisher Ratio (MSFR) [9] emerged as a response to the fundamental limitations in Phase Retrieval (PR) from spectral magnitude maps [10] under extremely low signal-to-noise ratio (SNR) [11] conditions. In laser spectroscopy and ultrafast electronic sensing, the recovery of the phase information from power spectral density (PSD) is a classic ill-posed problem, where the objective function often lacks a unique or stable solution due to the dominance of stochastic noise.

Our search for a solution led us to investigate the Singularity Complexity of the signal. We realized that while the energy (Shannon Entropy) might be lost in the noise, the "sharpness" of the information's intent (Fisher Information) remains as a latent trace. The synthesis of the MSFR was achieved by merging four distinct mathematical schools to regularize this ill-posed problem (Fig. 1):

- The Kolmogorov–Shannon Baseline: We utilize local Shannon entropy to quantify the global uncertainty of the "Empty-sky."
- The Skorokhod Singularity Mapping: To prevent over-smoothing, we adopt jump-process logic [12]. Instead of seeking a smooth phase recovery, we seek singularities—the points where the Fisher information spikes, signaling a deterministic "jump" from noise to mode.
- The Pastur Spectral Constraint: To distinguish between real signals and phantoms, we apply random matrix limits. The MSFR uses the Pastur distribution as a dynamic threshold: anything within the distribution is noise.
- The Glushkov Adaptive Control: We implement cybernetic feedback loop where the MSFR acts as the objective function, allowing the system to autonomously "lock" onto the phase of the ordered emission.

MATHEMATICAL FRAMEWORK

The central scientific contribution of this study is the development of the Modified Shannon-Fisher Ratio (MSFR) [13-14]. We transform the abstract information-theoretic duality into a practical computational tool for signal segmentation in extreme stochastic environments.

$$W_{(i,j)} = \frac{H_{\text{local}}(I_{(i,j)})}{H_{\text{local}}(\Delta I_{(i,j)}) + \varepsilon}$$

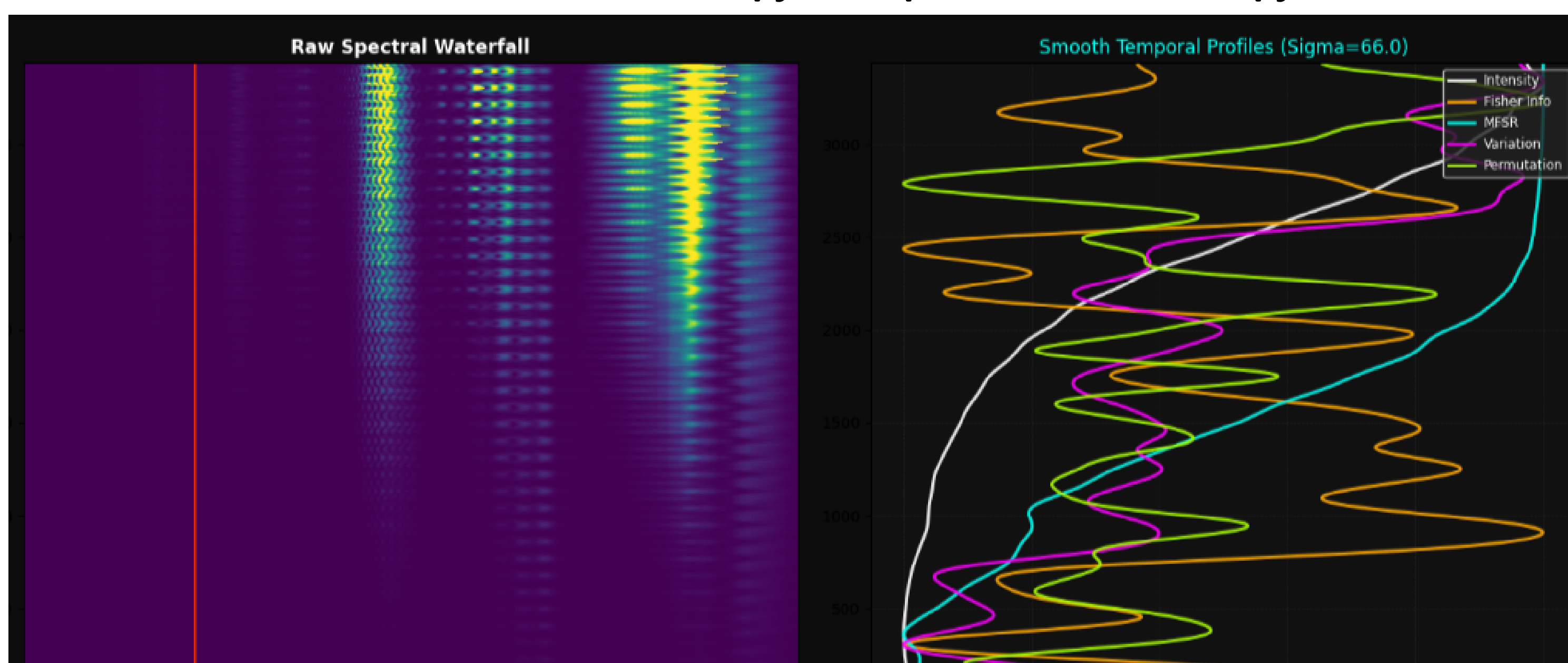
where:

$H_{\text{local}}(I_{i,j})$ Shannon entropy of spectral-temporal segment.

$H_{\text{local}}(\Delta I_{i,j})$ entropy of the derivative, serving as a proxy for Fisher information ε is a stabilization constant.

MSFR INDEX COMPARISON

Fig. 1. Comparative illustration of Average Modal Power, Fisher information, MSFR, variational entropy, and permutation entropy.



INNOVATION: THE CHAOS MAP

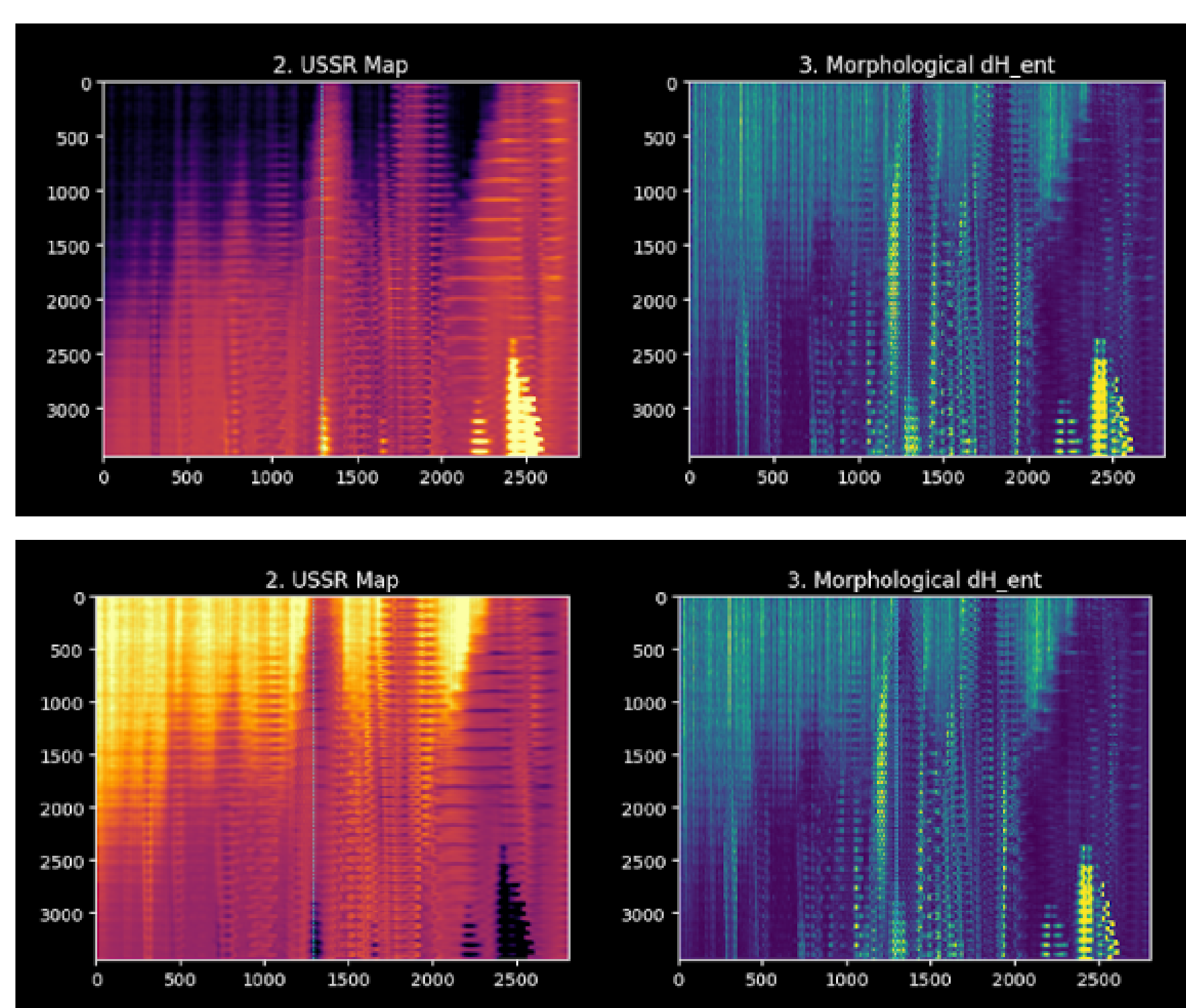
MSFR acts as a microstructure detector, suppressing stochastic noise and enhancing deterministic patterns. The result is a Chaos Map and an entropy-weighting mask (W_{entropy}).

$$W_{\text{entropy}}(i,j) = \exp(-\alpha \cdot W_{(i,j)})$$

where:

α is a scaling factor optimized experimentally. This mask performs a "singularity recall," enabling the system to suppress high-entropy noise and focus on ordered modal formations.

CHAOS MAP - MSFR GRADIENT



MORPHOLOGICAL GRADIENT ANALYSIS

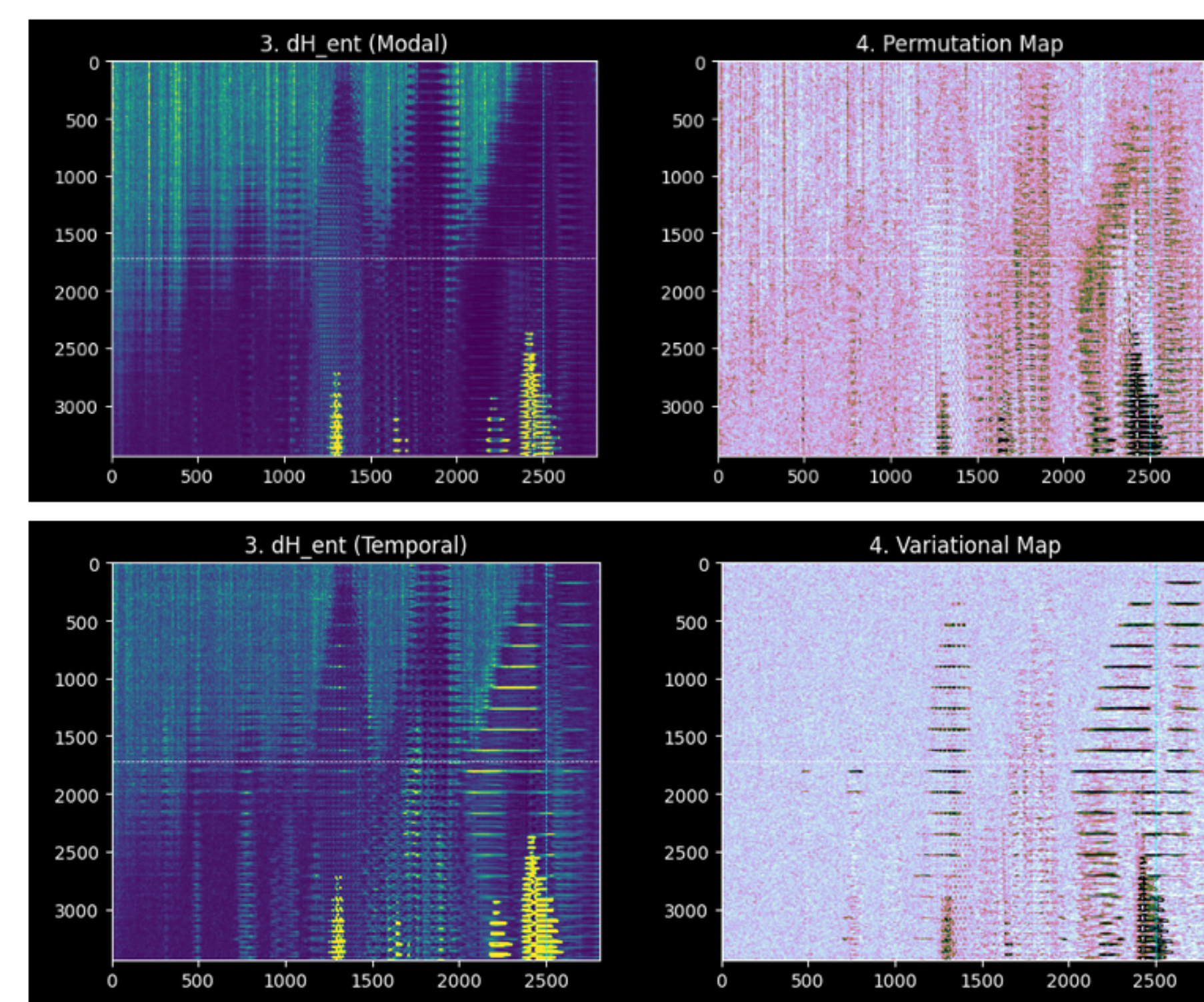
FROM ENTROPY FOAM TO MODAL SKELETON

The transition from global uncertainty to local structure is captured by shifting the entropy derivative from the temporal domain to the spectral (vertical) axis (ω) [15-17]. This allows the MSFR index to decode the internal "anatomy" of modal formations by evaluating the spectral morphological gradient (dH_{entropy}). The vertical gradient is computed using a discrete central difference of the localized entropy:

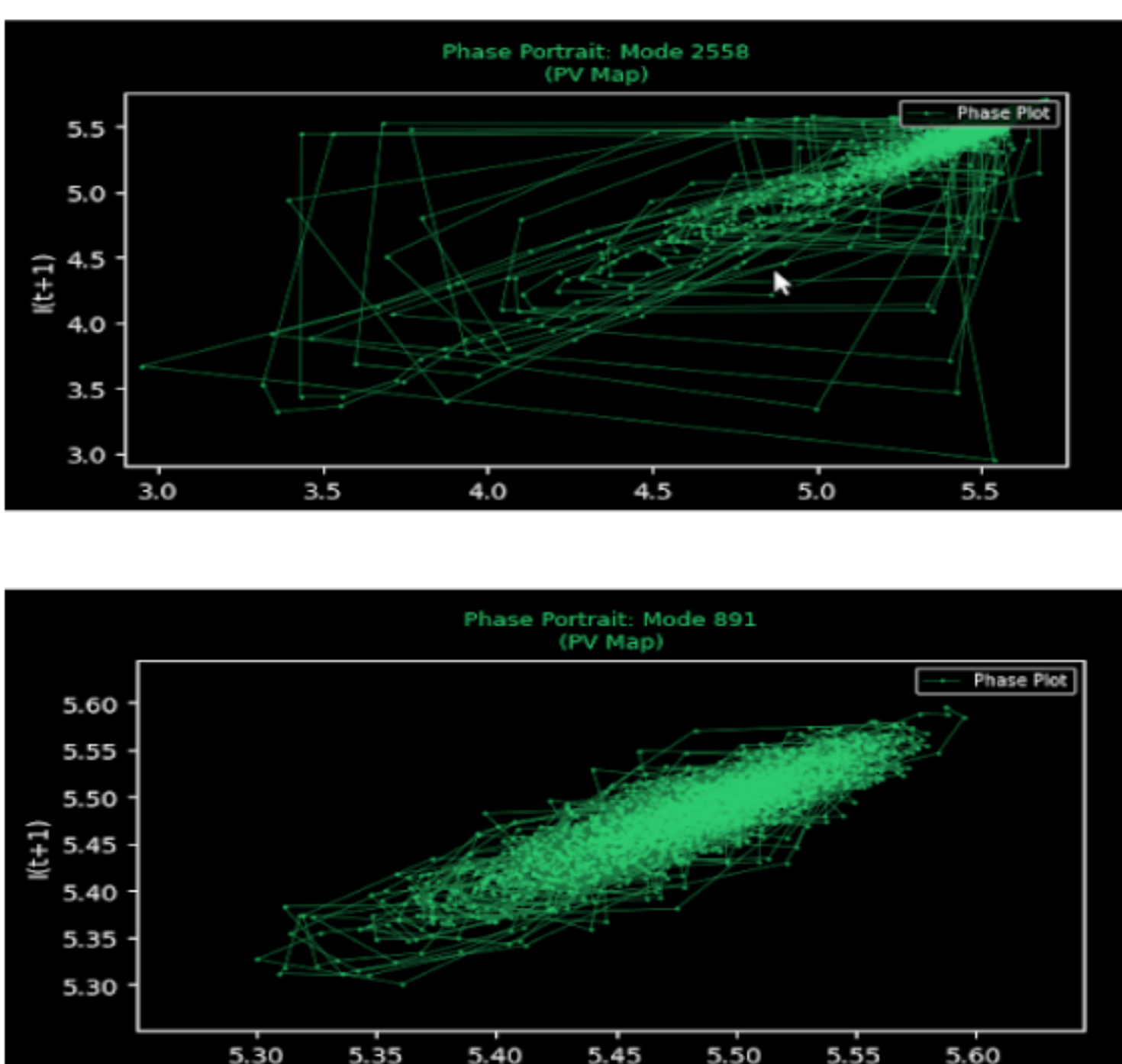
$$\partial H_{\text{entropy}}(\omega) = \frac{H_{\text{entropy}}(\omega + 1) - H_{\text{entropy}}(\omega - 1)}{2}$$

where: the formulation allows computation of both the inter-modal and temporal modal gradient, enabling analysis of structure evolution across frequency and time.

PERMUTATION-VARIATIONAL ENTROPY



MODAL PHASE PORTRAIT



The phase portrait serves as a topological representation of the system dynamics. In the absence of emission, the portrait remains compact, stochastic regime with no stable structural encoding. As modal emission emerges, the phase portrait expands and organizes into distributed trajectories, corresponding to coherent dynamical structures. This geometric transition can be interpreted as an increase of Kolmogorov complexity, where the system evolves. From an information-theoretic perspective, this process is analogous to a shortening of non-deterministic segments in a state-based representation: stochastic trajectories are replaced by localized, structured transitions. This behavior resembles Skorokhod-type jump localization, where discontinuities concentrate the system dynamics, and aligns with the idea of minimal description length associated with Glushkov-type representations. The observed expansion of phase-space volume (Fig. left) therefore reflects not an increase in randomness, but a reorganization into a structured, low-complexity regime governed by an efficient informational code. By detecting this transformation prior to intensity peaks, the system identifies deterministic precursors within the noise floor, indicating that the modal "program" is established before macroscopic emission becomes visible.

CONCLUSIONS

This study demonstrates that the transition from stochastic chaos to ordered emission in cristal structures can be quantitatively defined using the Modified Shannon-Fisher Ratio.

- **Topological Diagnostics:** Phase-space reveals the transition from stochastic confinement to structured modal dynamics.
- **Instability Differentiation:** Distinguishes noise-driven fluctuations from deterministic modal precursors.
- **Real-Time Recall:** Enables early detection of coherent structures before macroscopic emission.

MSFR is robust tool [18-20] for adaptive modal detection in high-entropy environments.

References

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